LandEmu v1: a learned emulator of the trait-based, individual-tree vegetation model LPJmL-FIT

Daily carbon and water exchange and forest structure at a fraction of the cost

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Abstract

Individual-based, trait-based vegetation models resolve how forests of differently adapted trees grow, compete and die, but they are too expensive to run inside Earth system models. We present LandEmu v1, a learned emulator of the dynamic global vegetation model LPJmL-FIT that reproduces both its daily carbon and water exchange and its yearly forest structure and trait composition, driven by daily weather alone. Its design follows from a systematic review of 115 emulator and hybrid-model studies of land models and from measurements that ruled out an earlier hybrid design: re-implementing the original's daily physics made it 4.6 times slower, and no single process of the original takes enough run time to give even a twofold speed-up. A five-member ensemble of small neural networks therefore replaces the entire daily step, with snow, top-metre and deep soil water carried as explicit stores that close the water balance exactly. Once a year, gradient-boosted tree models predict tree density, biomass per tree, four trait medians, leaf area, vegetation carbon and plant-type cover as a departure from a per-cell anchor, driven by the trailing 30-year climate; this part is not recursive, so its errors cannot accumulate. Tested on a climate model and an independent run of LPJmL-FIT withheld from their training, the yearly maps' error in dense forests is 1.26 (tree density) and 1.18 (biomass per tree) times that of a further run of the stochastic original, and smaller for the trait medians. Tested on cells withheld from their training, under climate models they had seen, the daily networks' 30-year mean fluxes are 2.4 (gross primary production) to 5.2 (runoff) times further from the original than a further run is — about 3–6 % per cell for production and evapotranspiration, more for runoff — with area totals within 3 %. After the start state has faded the structure error grows only slowly, with a creep in tree density under strong warming, and the flux error grows only for gross production under the two warmest climates. For a nitrogen-limited configuration of LPJmL-FIT with rising CO_2, the emulator reproduces 81–97 % of the change in vegetation carbon under two unseen climate models (a second run of the original: 61–89 %), but damps the largest warming-only loss (57–72 % of it) and the cell-to-cell spread of the response; the pre-registered test was not passed. The emulator costs 7.6×10^{-3} core-seconds per cell-year, about 35 and 900 times less than LPJmL-FIT at 25 and 500 patches.

1 Introduction

Terrestrial ecosystems take up about a third of anthropogenic CO_2 emissions (Friedlingstein et al., 2023), and the vegetation that does so is not a fixed surface but a population of plants that grow, compete, die and are replaced. Dynamic global vegetation models (DGVMs) represent this population in Earth system models (ESMs), usually through a few plant functional types with fixed properties. A growing body of work argues that this is not enough: the size structure and the functional diversity of forests shape how they respond to drought and warming, and models

that resolve individuals, cohorts and traits behave differently from those that do not (Fisher et al., 2018; Fisher and Koven, 2020). LPJmL-FIT, a version of the LPJmL model with individual trees and variable traits (Sakschewski et al., 2015, 2016; Thonicke et al., 2020), is one of these models: each tree carries its own specific leaf area, wood density, rooting depth and drought threshold, drawn when it establishes, and the forest's trait composition shifts as trees with poorly suited traits die. Amazon forests with more trait diversity recover better from drought in this model than forests without it (Sakschewski et al., 2016).

The price of this realism is cost. LPJmL-FIT simulates every tree every day, and because establishment and mortality are random it averages over many replicate patches per grid cell: 25 in the runs used here, around 500 in published applications. Its run time grows in proportion to the number of patches, which puts it orders of magnitude beyond what an ESM can afford for its land surface. Emulation — replacing the expensive model with a fast statistical or machine-learned approximation — is the obvious route, and the land-modelling literature contains many emulators. A systematic review of 115 emulator and hybrid studies of DGVMs and land surface models (Priesner, 2026) shows, however, that most of them serve calibration and sensitivity analysis: 89 emulate only one or two variables, a single domain or a single site, or a summary statistic. Only seven emulate a land model broadly enough to stand in for part of it, and each covers only part of the model: prognostic soil states and turbulent fluxes of a weather-prediction land scheme (Wesselkamp et al., 2025; Raoult et al., 2026), carbon and stand structure of a demographic model (Wang et al., 2023; Li et al., 2026), biome-level carbon for integrated assessment (Ekholm et al., 2024), equilibrium carbon pools for spin-up (Gao et al., 2025) or steady-state parameter responses (Ricciuto et al., 2018). None emulates size or trait distributions or individual mortality; the closest is a state-transition network for the landscape model iLand (Rammer and Seidl, 2019). None of the seven has been coupled to an ESM, only 12 of the 115 studies test climates outside the training range, and only 25 check conservation.

The review also distils what goes wrong, and these lessons shaped the emulator described here. Tree ensembles trained to predict one step ahead drift when run freely at long lead times (up to about a year in the cited tests), while networks trained on their own multi-step rollouts do not (Wesselkamp et al., 2025; Raoult et al., 2026). Agreement in the present climate says little about agreement under a changed climate: an emulator of the ecosystem model ecosys matched present-day net primary production almost exactly yet underestimated it by $204\,\mathrm{gC\,m^{-2}\,yr^{-1}}$ by 2100 (Shirley et al., 2023), and an LPJ-GUESS emulator trained on stylised climate perturbations failed to transfer to realistic scenarios (Natel et al., 2025). Offline skill can collapse once feedbacks act, as for a machine-learned fire model in a land surface model whose spatial correlation fell from 0.96–0.98 to 0.59–0.70 (Liu et al., 2024, 2025). Water and carbon budgets should be closed by construction rather than encouraged by a loss term (Baghirov et al., 2026). From this evidence the review derives six requirements for emulators that are to be used inside ESMs: prognostic state and memory, closure of the coupling interface, conservation, stability under feedback, credibility under out-of-distribution climate, and quantified uncertainty with usable software interfaces.

This paper describes LandEmu v1, an emulator of LPJmL-FIT built against these requirements. Its contributions are:

1. a complete emulator of an individual-tree, trait-based DGVM (Fig. 1) — daily gross and net primary production, interception, transpiration, soil evaporation, runoff, snow and soil water, together with yearly tree density, biomass per tree, four trait medians, leaf area, vegetation carbon and plant-type cover — that runs from daily weather alone and conserves water exactly;
2. an account of how its design followed from measurements that ruled out a hybrid design (Sect. 3);
3. an evaluation that holds out a whole climate model and an independent run of the original (yearly maps) or whole blocks of cells (daily networks), and that judges every error against the original's own run-to-run variability (Sect. 5);
4. a test of whether the emulator learns the warming response when the training data cover the test climate, and a full-grid application (Sect. 6);

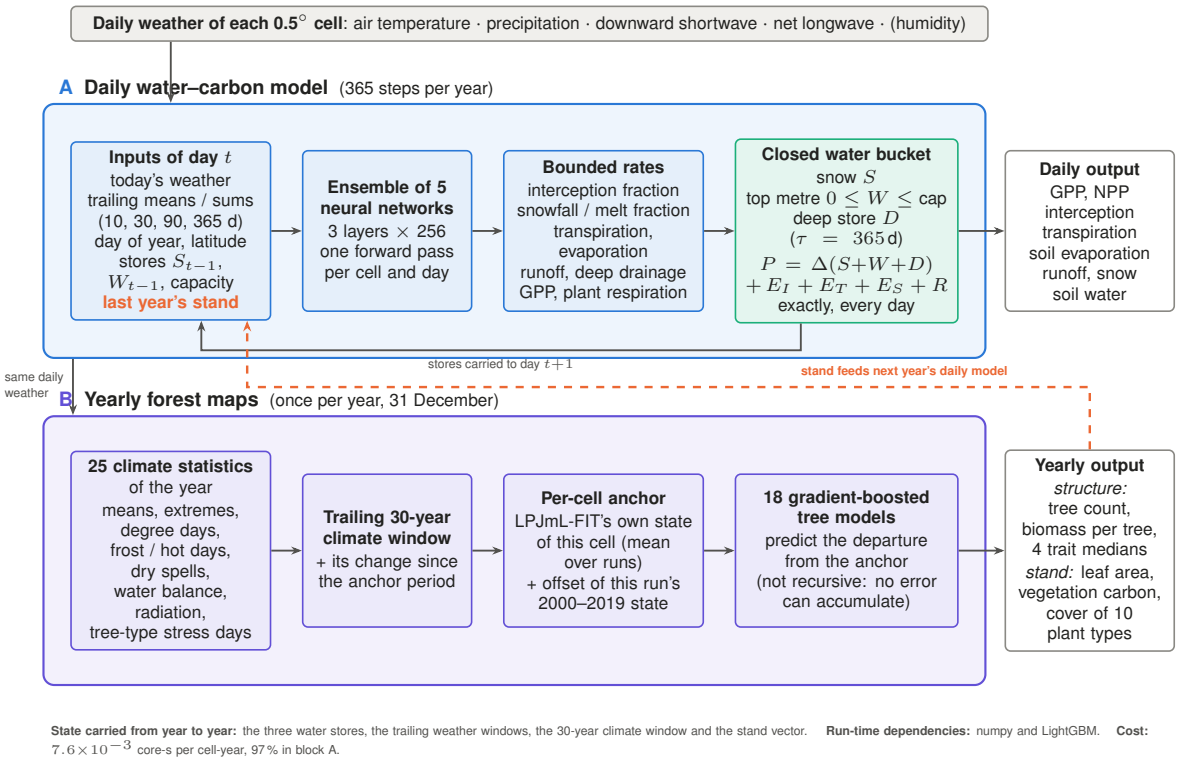


Figure 1. How LandEmu advances one year. (A) The daily water–carbon model: the inputs of day t pass through an ensemble of five neural networks whose bounded outputs drive a closed three-store water bucket; the stores are carried to the next day. (B) The yearly forest maps: 25 climate statistics of the year enter a trailing 30-year window, and 18 gradient-boosted tree models predict the forest structure and stand as a departure from a per-cell anchor. The stand (leaf area, vegetation carbon, cover of ten plant types) is the daily model's input in the following year (dashed). Every other quantity is predicted afresh each year.

5. an end-to-end cost measurement on one core against the original at its training and publication settings.

2 The model to be emulated: LPJmL-FIT

LPJmL-FIT couples the water, carbon and vegetation dynamics of LPJmL (Schaphoff et al., 2018) with an individual-tree forest module (Sakschewski et al., 2015; Thonicke et al., 2020). On each 0.5° grid cell the forest grows on a set of replicate patches (225 m^2 each). Trees of seven woody plant types establish with traits drawn within type-specific ranges, or inherited with small variations from large trees of the same patch; they compete for light in a layered canopy and for water in a 23-layer soil, allocate carbon each year, and die from carbon starvation, age, heat and drought stress. Three grass types fill the remaining space. The daily step computes photosynthesis and stomatal conductance per tree, interception, transpiration, soil evaporation, snow, infiltration, percolation and runoff; the yearly step allocates carbon, applies mortality and establishes new trees. We use the configuration with natural vegetation only. For most of this paper nitrogen limitation is switched off and atmospheric CO_2 is held constant in the future runs (at its 2019 level; 2014 in the runs of Sect. 6.6); this is how LPJmL-FIT is usually run, because without nitrogen limitation its CO_2 fertilisation is unbounded. The emulator of this configuration therefore takes no CO_2 input, which is faithful to the original rather than a gap. Section 6.5 uses a second configuration, with nitrogen limitation (von Bloh et al., 2018) and rising CO_2, for which the emulator receives CO_2 as an input.

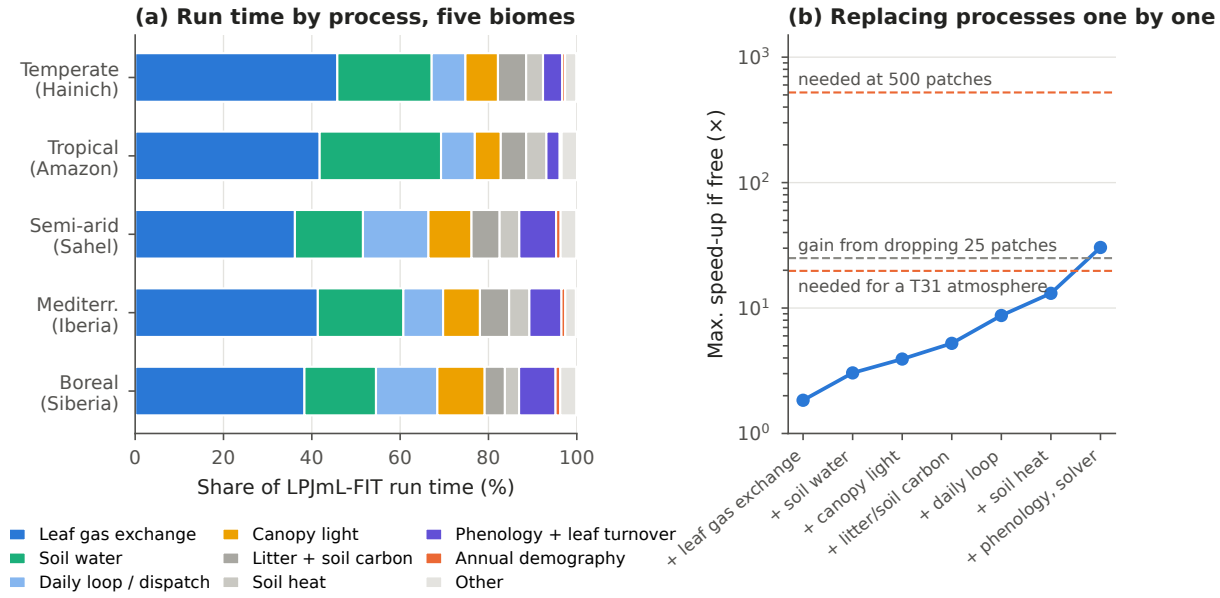


Figure 2. Where LPJmL-FIT spends its time. (a) Exclusive run time by process at five biome cells (25 patches, one core; mathematical library calls attributed to their callers). (b) The largest speed-up obtainable at the temperate cell if processes were made free of cost one after another, against the speed-up needed for an intermediate-resolution (T31) atmosphere, the gain from dropping the 25-patch ensemble, and the speed-up needed at 500 patches.

Where the run time goes. We profiled the unmodified production binary at five contrasting cells (temperate, tropical, semi-arid, Mediterranean, boreal; 25 patches, one core) and attributed the exclusive run time to processes, assigning the statically linked mathematical library calls (21–27 % of all time) to the routines that call them (Fig. 2a). Leaf gas exchange takes 36–46 % of the run time and soil water another 15–27 %. The yearly demography — allocation, mortality, establishment and turnover — takes only 0.4–1.1 %, and the daily loop as a whole 97–98 %. The marginal cost is 0.20–0.33 core-seconds per cell-year at 25 patches, and it grows linearly with the number of patches. Even if the eight most expensive processes were free of cost, the model would speed up at most 30-fold at the temperate cell (Fig. 2b), whereas an intermediate-resolution atmosphere leaves an allowance of about 0.0135–0.030 core-seconds per land cell-year (10 % of a measured coupled atmosphere's cost at T31 and T63 resolution). Removing the 25-patch ensemble alone is worth a factor of 25, and of about 500 at the publication setting.

The original is stochastic. Two independent runs of LPJmL-FIT — each with its own 1000-year spin-up and random seed, under identical climate — disagree on a typical cell by about 6 % in the number of trees taller than 5 m, 12 % in biomass per tree, 2–10 % in the trait medians and 1.0–1.6 % in 30-year mean fluxes. In sparse stands the disagreement is far larger. Any emulator can at best predict the expectation over such runs; this sets both the target and the yardstick of the evaluation (Sect. 5).

3 From a hybrid to a fully learned emulator

LandEmu is the second architecture we built for LPJmL-FIT. The first was a hybrid, the design favoured by much of the hybrid-modelling literature (Reichstein et al., 2019; Shen et al., 2023): learn the slow, poorly constrained part and keep the physics. It had three components (Fig. 3, left). A learned yearly demography predicted the number of trees per patch with a distributional random forest, sampled new trees' traits from a Gaussian copula, and applied the original's mortality rules, ported from its source code. The daily physics — photosynthesis, the stomatal solve, the layered

canopy, soil water, phenology, allocation and growth — was re-implemented as differentiable code, validated process by process against the original at five biome cells. An energy closure supplied the skin temperature and sensible heat that LPJmL-FIT does not compute. The training data were two runs of the original under one historical and one future climate. Five findings — four measurements and one from the literature — ruled this design out (Fig. 3, middle).

(i) It was not faster. The differentiable re-implementation cost 1.23 core-seconds per cell-year at 25 patches, 4.6 times more than the original. Per tree, its daily step cost 51–58 times the original's (83 % of it in the iterative stomatal solve with a finite-difference derivative); only because it carried an order of magnitude fewer trees (those taller than 5 m) was the overall factor as small as 4.6. Measured solver improvements would have reached parity at best.

(ii) The original's cost is the daily step and the patch ensemble (Sect. 2). Any substantial speed-up must replace the daily step as a whole and must predict the expectation over patches instead of simulating replicates.

(iii) One-step learning drifts when run freely. The tree-count model reached an out-of-sample coefficient of determination of 0.982 one year ahead — but carrying the previous year's count forward already reached 0.962. Fed its own output, its area-weighted warming response turned from +0.71 to −0.23 of the original's within decades, because it followed 96 % of large increases but only 87 % of large decreases: a small, state-dependent asymmetry that accumulates. This is the failure the review documents for one-step tree ensembles (Wesselkamp et al., 2025).

(iv) One climate per place cannot teach a response. Across the global grid, 76 % of a cell's warming is predictable from its present climate, so with one historical and one future run the climate response is confounded with geography — the identification problem behind the out-of-distribution failures in the literature (Shirley et al., 2023; Natel et al., 2025).

(v) The literature points the same way: conservation by construction, out-of-range climate tests and explicit state are what the published emulators most often lack (Priesner, 2026).

From these findings follow five design principles (Fig. 3, right). **P1**: learn the whole daily step with one small network per cell and day, keeping water stores explicit and closed. **P2**: predict the expectation over the patch ensemble, so that the number of patches drops out of the cost. **P3**: do not make the forest state recursive; predict each year's state as a departure from a per-cell anchor, driven by climate, so that errors cannot accumulate. **P4**: train on climate contrast — many climate models and scenarios and several independent runs — and test on a climate model and a run never seen. **P5**: measure every error against the original's own run-to-run variability.

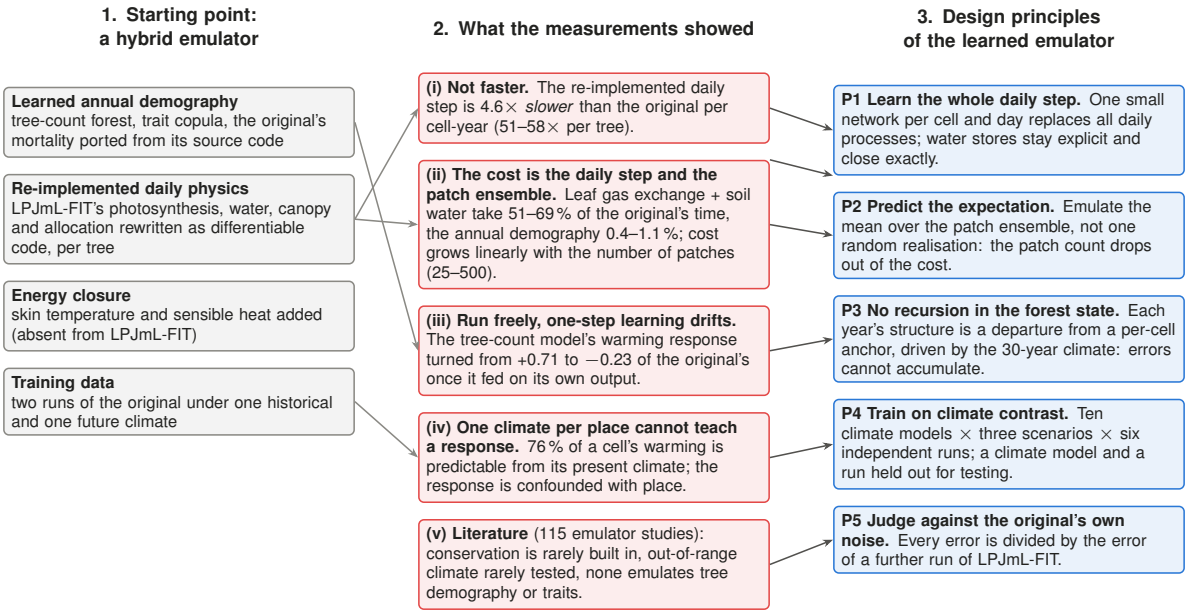


Figure 3. How the design was arrived at. Left: the components of the first, hybrid emulator. Middle: five findings that ruled it out, each a measurement on LPJmL-FIT or on the hybrid, or a result of the literature review. Right: the design principles of LandEmu that answer them.

Table 1. What LandEmu reads, carries and returns.

Daily input	air temperature, precipitation, downward shortwave, net longwave (all days); specific humidity (yearly statistics only); CO_2 in the nitrogen-on variant
Per-cell set-up	latitude, longitude, soil code, top-metre plant-available water capacity; anchors of every yearly quantity; the climate of the anchor runs; the run's own 2000–2019 state and its offset from the anchor; initial stores, stand and weather windows
State carried	snow, top-metre water, deep water; trailing weather sums and means (10, 30, 90, 365 days); 30-year window of yearly climate statistics; last year's stand
Daily output	gross primary production (GPP), net primary production (NPP), interception, transpiration, soil evaporation, runoff, snow water equivalent, top-metre soil water, deep store
Yearly output	trees taller than 5 m per patch, above-ground biomass per tree, medians of specific leaf area, wood density, maximum rooting depth and drought threshold; stand leaf area index, vegetation carbon, fractional cover of ten plant types

4 Emulator description

LandEmu advances one year at a time in two blocks (Fig. 1; Table 1). Block A, the daily water–carbon model, runs 365 daily steps from the day's weather and last year's stand. Block B, the yearly forest maps, runs once at the end of the year from climate statistics of the same weather and predicts the forest structure and the stand, which becomes the daily model's input for the next year. Everything is driven by daily weather on the 0.5° grid: air temperature, precipitation, downward shortwave radiation and net longwave radiation; specific humidity enters only the yearly climate statistics. All years have 365 days.

4.1 Daily water–carbon model

Inputs. For cell c and day t the network sees today's weather; trailing means of temperature (10 and 30 days) and shortwave radiation (30 days) and trailing sums of precipitation (30, 90 and 365 days), which stand in for the memory of the deep soil without carrying it as a state; the day of year (as sine and cosine), the sine of latitude and the day length; yesterday's snow S_{t-1} and

top-metre water W_{t-1}, W_{t-1} divided by the capacity, and the capacity itself; and the stand at the end of the previous year (leaf area index, vegetation carbon and the cover of the ten plant types). No input is taken from the fluxes of the run being emulated. The top-metre water capacity, which the original recomputes from its evolving soil carbon, is read from the original each year in the evaluation runs; it changes over the century by a median 0.1–0.2 % (at most 4–6 % in single cells), so a fixed per-cell value would serve.

Network. Each of five ensemble members is a multilayer perceptron with three hidden layers of 256 units and SiLU activations; all outputs share one trunk. The outputs are mapped to bounded rates: an interception fraction $i \in [0, 1]$, a snowfall fraction $f \in [0, 1]$ and a melt fraction $m \in [0, 1]$, non-negative transpiration E_T, soil evaporation E_S and runoff R, a signed exchange Q between the top metre and the deep soil, non-negative GPP and plant respiration R_a, so that NPP = GPP − R_a. The ensemble mean of the five members is used.

Closed water bucket. With precipitation P_t, the stores are updated as

$$S_t = S_{t-1}(1 - m) + fP_t, \qquad E_{I,t} = iP_t, \tag{1}$$

$$W_t = W_{t-1} + P_t - (S_t - S_{t-1}) - E_{I,t} - E_{T,t} - E_{S,t} - R_t - Q_t, \tag{2}$$

$$D_t = (D_{t-1} + Q_t)\,(1 - 1/\tau)\,, \tag{3}$$

where D is a deep store that drains the amount $(D_{t-1} + Q_t)/\tau$ into runoff each day, with a time constant $\tau = 365$ days. Water above the top-metre capacity is added to runoff and a deficit below zero is taken from the deep store. Hence, every day and to rounding precision,

$$P_t = \Delta(S + W + D)_t + E_{I,t} + E_{T,t} + E_{S,t} + R_t^{\mathrm{tot}}, \tag{4}$$

with R^{tot} the total runoff. The bounds restrict nothing the original does: on eight million cell-days interception and snow accumulation never exceeded precipitation and no flux turned negative. Because the deep store can be drawn below zero, its drainage term, and with it total runoff, turns slightly negative on a few days (minimum −2.5 to −10 mm per day in the cells tested). The deep store's drainage was added after a first version let it accumulate 2–3 m of water over 81 years, ten times the original's change.

Training. The networks are trained one step ahead (yesterday's stores from the original) on 40 million sampled cell-days of 730 training cells, with 80 further cells for early stopping, minimising the sum over the eight targets of the squared errors standardised by each target's spread (Adam with a one-cycle learning-rate schedule peaking at 3×10^{-3}, batches of 16 384, ten epochs; about 15 minutes per member on one GPU). They are then run freely: the stores are carried by the model over 81 years. A free run whose error grew to twice the one-step error would have triggered rollout training; it never did. Specific humidity was dropped as an input because the original's daily exchange does not read it; removing it changed no score beyond run-to-run noise.

4.2 Yearly climate statistics

At the end of each year, LandEmu computes 25 statistics of the year's daily weather: mean, coldest-month and warmest-month temperature; degree days above 5 °C; frost days and days above 30 °C; annual and summer precipitation; potential evaporation; the annual, summer and driest three-month climatic water balance; the longest dry spell; mean relative humidity; summer vapour pressure deficit and its 10-day extremes; mean shortwave and net longwave radiation; and, for each of the seven tree types, the number of days outside its temperature-stress range. These enter a trailing 30-year window, whose means are the climate the forest maps see.

4.3 Yearly forest maps

For each yearly quantity q (six structure quantities and twelve stand quantities) a separate gradient-boosted tree model (Ke et al., 2017) predicts

$$q_{c,y} = A^q_{c,y} + g_q\Big(\bar{x}_{c,y},\ \bar{x}_{c,y} - \bar{x}^A_{c,y},\ A_{c,y},\ s^0_c,\ \Delta s^0_c,\ \phi_c, \lambda_c, \sigma_c\Big)\,, \tag{5}$$

where $A^q_{c,y}$ is the *anchor*: the original's mean of q at cell c in year y over its training runs and climates; $\bar{x}_{c,y}$ is the trailing 30-year climate and $\bar{x}^A_{c,y}$ the corresponding climate of the anchor runs, so that their difference measures how far this climate departs from what the anchor represents; $A_{c,y}$ collects all anchors of the cell; s^0_c is the run's own 2000–2019 state and Δs^0_c its offset from the training runs' mean (the known start state); and ϕ, λ, σ are latitude, longitude and soil code. Biomass per tree is predicted as a logarithmic ratio to its anchor, which weights relative errors equally across dense and sparse stands; the other quantities as differences. Each model has 400 trees with 31 leaves, learning rate 0.05 and 80 % row and column subsampling, and is trained on every fourth year of the training runs: five runs under nine climate models in the test folds (about three million cell-years), all six runs under all ten climate models in the deployment release (about four million).

Two properties of this form matter. First, the map is not recursive: no prediction is fed back into the forest maps, so a year's error cannot carry into the next (P3). The only quantity passed from the maps to the daily model is the stand, which affects next year's fluxes but not next year's maps. Second, the anchor carries what is specific to a cell — its soil, its history, the composition the original grew there — so the learned part only has to explain how a cell departs from it under a different climate. The price is that the anchor must exist: the original must have been run on every cell the emulator is to run on (Sect. 7.3). For the full-grid application (Sect. 6.6) the anchor is instead the cell's own simulated 1985–2014 state, constant in time.

4.4 Implementation

LandEmu is a Python package whose only run-time dependencies are numpy and LightGBM; the networks are ported from PyTorch to numpy (maximum difference 10^{-6} of each output's range on 20 000 real input rows). The yearly climate statistics of the package equal those of the training pipeline exactly, and a validation release reproduces the scored daily fluxes of the evaluation to rounding (median per-cell difference of 30-year means 1.6–2.9$\times10^{-4}$) and the yearly quantities with a median difference of zero (99th percentile 0.1–7 %). The remaining yearly differences have one cause: the training table converted the forcing, stored in 0.1 °C steps, in single precision, so days at exactly 30.0 °C counted as hot days; continuous forcing from a host model never sits exactly on such a threshold. A release (models plus per-cell set-ups for 1 050 cells) occupies 183 MB. The interface offers a whole year (`step_year`) or a day at a time (`begin_year`, 365 × `step_day`, `end_year`, bit-identical to the yearly call), so that a host model can exchange fluxes daily.

4.5 Variant with a CO_2 input

For LPJmL-FIT with nitrogen limitation and rising CO_2, the same architecture receives the year's CO_2 concentration as an additional daily-network input, and the forest maps receive the trailing 30-year mean CO_2 and its change since 1990–2019. It is trained on runs of that model (Sect. 6.5).

5 Training data and evaluation

5.1 Simulations of the original

The main data set is a panel of 1 050 cells: 105 blocks of ten contiguous 0.5° cells, chosen to balance 20 climate strata (Fig. 4a). Each block was spun up for 1000 years and run over

2000–2019 under observed weather (GSWP3-W5E5; Lange et al. 2021) and over 2020–2100 under bias-adjusted output (Lange, 2019) of ten CMIP6 climate models (GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0, UKESM1-0-LL, CanESM5, CNRM-CM6-1, CNRM-ESM2-1, EC-Earth3, MIROC6) for three scenarios (SSP1-2.6, SSP3-7.0, SSP5-8.5; O'Neill et al. 2016), plus a control with observed 1990–2019 years resampled. We call each combination of climate model and scenario a *leg*. Six independent runs exist, each with its own spin-up and random seed, all with 25 patches, nitrogen limitation off and CO_2 constant after 2019. Per-tree output is written every year (trees taller than 5 m, as in the original's standard output); daily fluxes are written for 13 legs of five climate models and for the control, in runs 1–4.

5.2 Held-out tests

The forest maps are tested in folds that hold out one of five climate models entirely, together with run 4: they are trained on the other nine climate models and runs 1, 2, 3, 5 and 6, and run 4 under the held-out model is the truth (Fig. 4b). With three scenarios this gives 15 test cases. The cells themselves are seen in training, as they would be in deployment; this choice is discussed in Sect. 7.3. The daily networks are trained on runs 1–4 under all 14 legs with daily output and tested on 24 blocks (240 cells, 238 with GPP of at least $50\,\mathrm{gC\,m^{-2}\,yr^{-1}}$) never used in their training, including the five biome blocks of Fig. 2. For them, therefore, only the place is new: they have seen every climate model they are scored on. In the full emulator the daily networks read the stand predicted by forest maps that never saw the test run or the test climate model. Earlier tests with a climate model withheld from the daily networks are summarised in Sect. 6.4.

5.3 Measuring error against the original's own variability

Because a second run of LPJmL-FIT differs from the first, a fixed tolerance such as "within 10 %" is uninformative: two runs of the original agree on all six structure quantities within 10 % (or within the typical two-run spread of the cell's density class, where that is larger) in only 17–21 % of cells. Following practice in climate-model emulation and model verification (Watson-Parris et al., 2022; Baker et al., 2015; Lütjens et al., 2025), we instead compare the distribution of errors over cells with that of a further run. For cell c and quantity q against the unseen truth run T, the emulator's error is $e_E(c) = |E(c) - T(c)|/|T(c)|$ and a further run R's error is $e_R(c) = |R(c) - T(c)|/|T(c)|$, pooled over the available further runs. The *error ratio* at centile k is

$$\rho_k = \frac{P_k\,[e_E]}{P_k\,[e_R]}, \tag{6}$$

with $k = 50$ (the typical cell) and $k = 90$ (the worst tenth of cells). A further run of the original scores $\rho \approx 1$ by construction; the mean of three runs scores about 0.82, and a perfect predictor of the expectation about 0.71, because an average beats a single random realisation. Values below one are therefore possible and, for a model that predicts the expectation, expected. We apply the ratio to 30-year means (2071–2100 for structure; 2021–2050 and 2071–2100 for fluxes), to the climate response (late minus early period), and to single years (for drift). Area totals (cosine-latitude weighted) are reported as signed relative errors, because a coherent bias of a few per cent passes every per-cell test yet does not average out in a sum.

Pass standard. Forests with at least five trees per patch ("dense"; about 80 % of tree-bearing cells) must have a typical error ratio of at most 1.2, a worst-tenth ratio of at most 1.5 and area totals within 5 %; sparse stands must stay within 3 times a further run and their totals within 25 %. A criterion passes when its median over cases meets the limit and at least 12 of 15 cases do. The standard as first written classed a cell as dense by the held-out run itself. That selects cells on the outcome being predicted — a cell whose truth run happens to lose trees moves into the sparse class — so we class cells by the anchor instead, before the run, and report the original definition

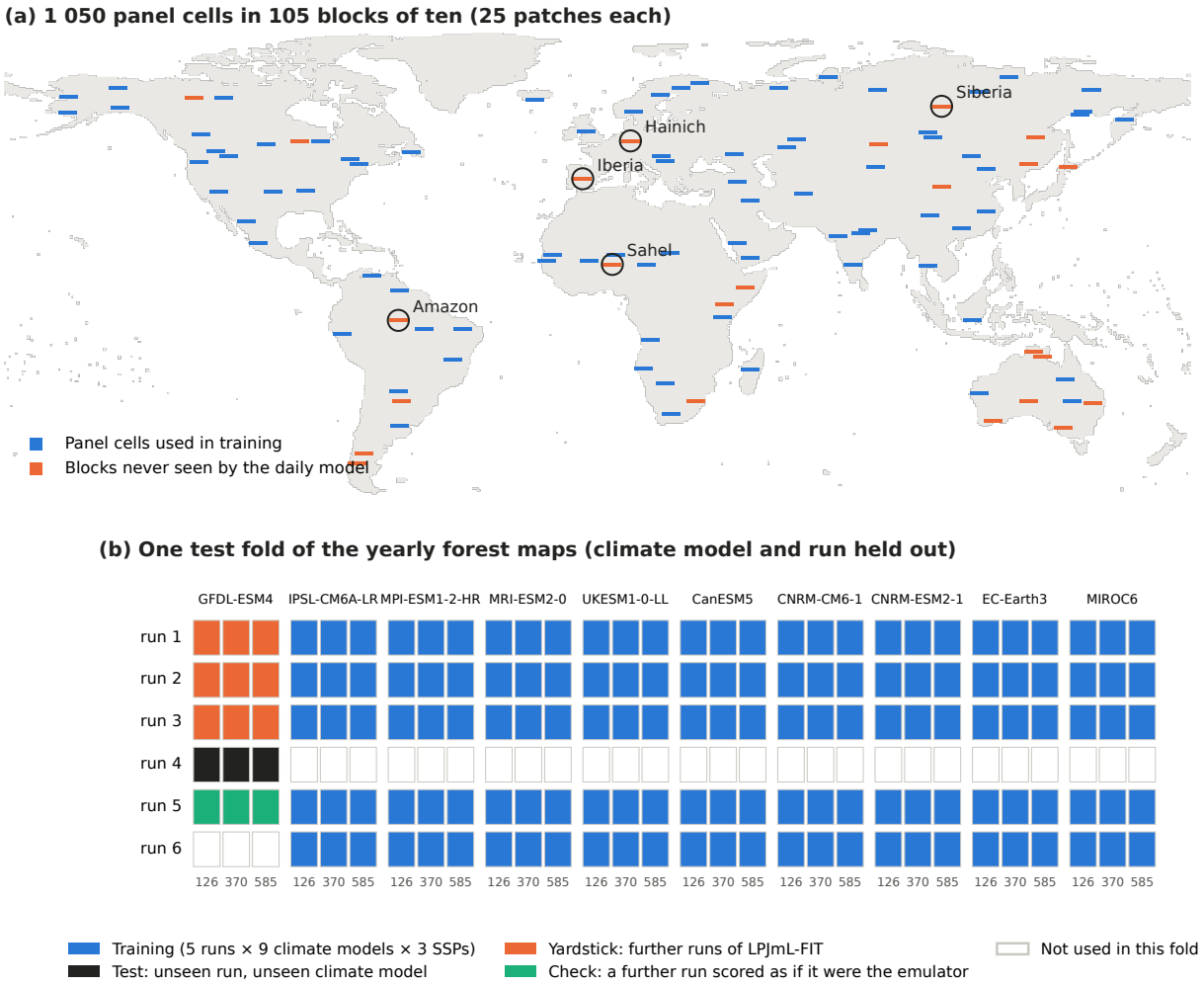


Figure 4. Training and test design. (a) The 1 050 panel cells (105 blocks of ten cells), the 24 blocks never used to train the daily networks, and the five biome blocks. (b) One of the five test folds of the yearly forest maps: one climate model (here GFDL-ESM4) and one independent run are held out; further runs under the held-out climate model give the yardstick, and one of them is scored as if it were the emulator to check the scorer.

beside it (Sect. 6.2). The daily fluxes need not be as close to the original as a second run is; for them we propose a typical and worst-tenth ratio of at most 3 (the sparse-stand limit above), a climate-response ratio of at most 1.5 and area totals within 5 %, numbers set after the first daily scores were known. *Drift* means the gap to the original growing, not the values changing; we measure it as the ratio of the single-year typical error ratio in 2081–2100 to that in 2021–2040, with a limit of 1.2. For the forest structure, whose early years benefit from the known start state, the pre-registered form of this rule failed for that reason, and we additionally report the ratio to 2051–2070, a baseline chosen after that result (Sect. 6.3). Because several of these choices were made with some scores known, we report every value so that readers can apply their own limits.

Reference arms. Every score is reported beside (i) a further run scored as if it were the emulator, which must give about 1 and checks the scorer; (ii) the mean of other runs, a reference for a near-ideal predictor of the expectation; (iii) the anchor alone, without the learned departure; and (iv) the run's 2019 state carried forward, which shows whether the measure can detect drift.

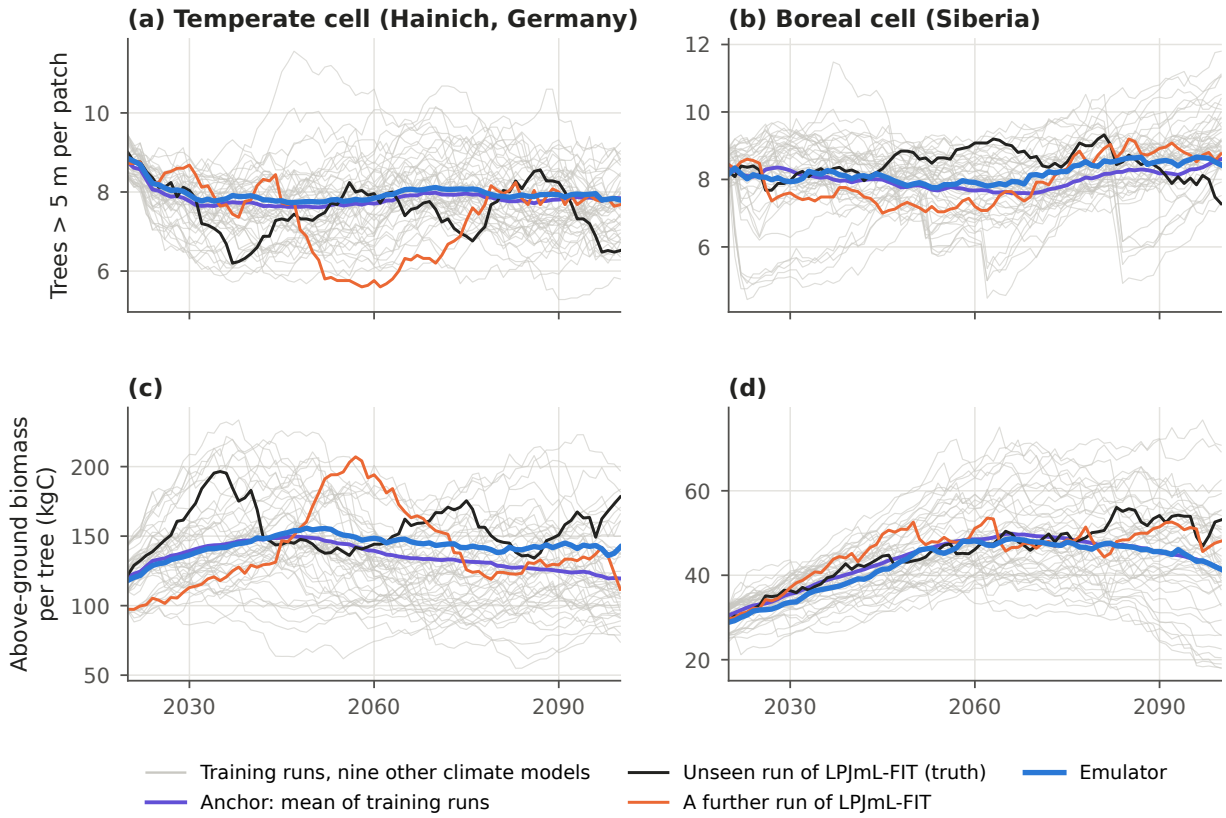


Figure 5. The yearly forest maps at two single cells under a held-out climate model (GFDL-ESM4, SSP3-7.0): (a, b) trees taller than 5 m per patch and (c, d) above-ground biomass per tree. Grey: training runs under the nine other climate models; violet: their mean, the anchor; black: the unseen run of LPJmL-FIT; orange: a further run under the same climate; blue: the emulator.

6 Results

6.1 How the forest maps behave at a single cell

Figure 5 shows what the yearly maps do. At the temperate and the boreal cell the original's runs scatter widely: its unseen run and a further run of the same climate wander apart by up to two trees per patch and by 50 % in biomass per tree for decades, because a few deaths or recruitments in 25 patches change the stand. The anchor — the mean of the training runs under other climates — smooths this noise but knows nothing about the held-out climate. The emulator follows the anchor where the climate resembles the training climates and departs from it where it does not: at the temperate cell it keeps biomass per tree above the declining anchor in the second half of the century, as the unseen run does. It does not reproduce the excursions of the single run, and it should not: they are not predictable from climate.

6.2 Forest structure

Over all 15 held-out cases (Fig. 6), the dense forests' typical error of tree density is 1.26 times that of a further run (limit 1.2; met in 5 of 15 cases, where 12 are required), and that of biomass per tree 1.18 times (met in 9 of 15). The four trait medians are closer to the unseen run than a further run is (0.59–0.86 times): the emulator predicts the expectation, while a further run is itself a random draw. In the worst tenth of dense cells tree density reaches 1.47 times (limit 1.5, met in 10 of 15 cases) and biomass per tree 1.10 times a further run. Area totals over dense cells are off by a median of −0.2 % for stems (cases −2.9 to +1.2 %) and −1.5 % for biomass (−7.6 to +4.4 %; three cases past the 5 % limit). Sparse stands are harder: tree density 3.00 times a further

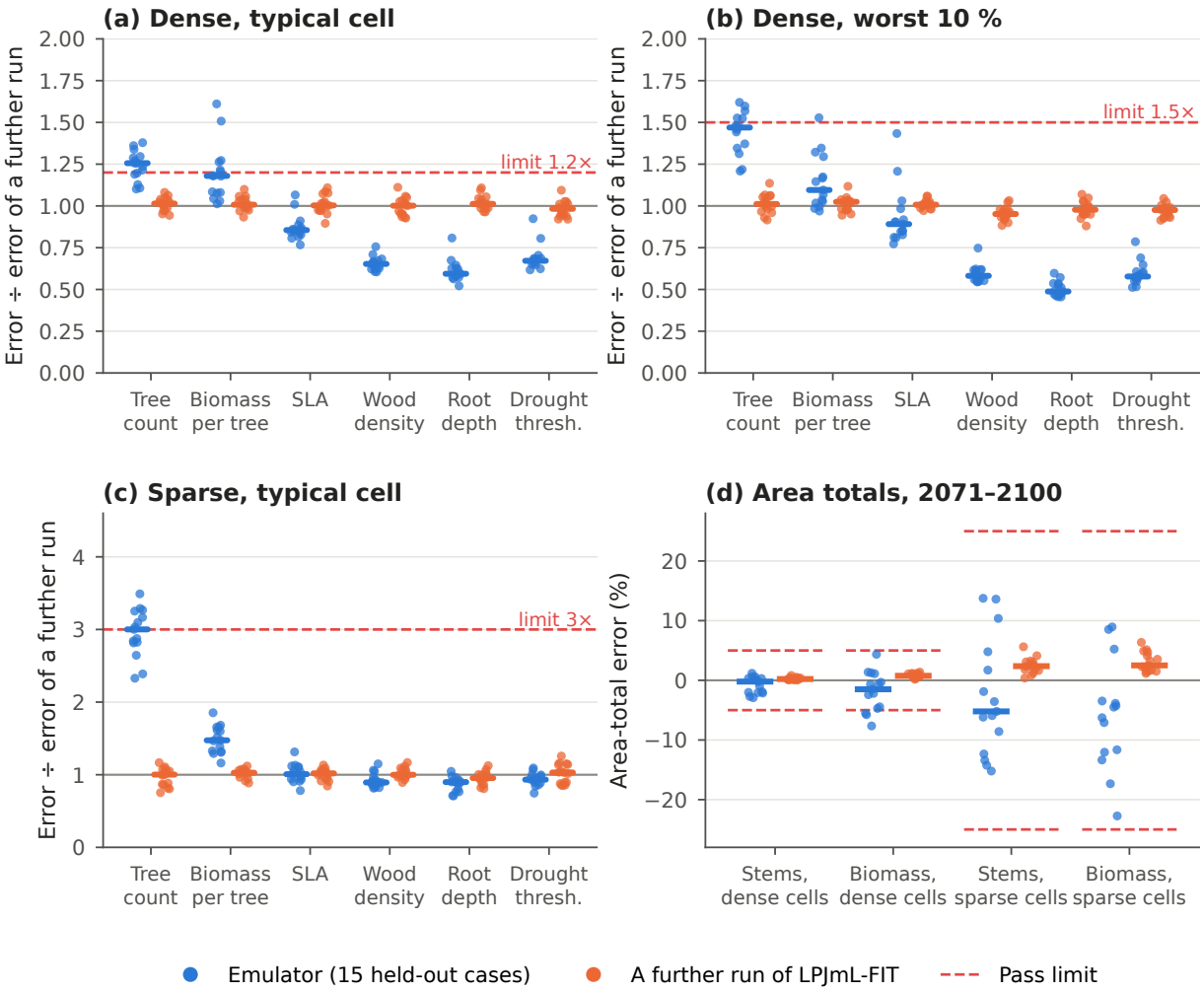


Figure 6. Forest structure in 2071–2100 under a held-out climate model and run (15 cases: five climate models × three scenarios). (a–c) Error ratio for six quantities: typical dense cell, worst tenth of dense cells and typical sparse cell; dots are cases, bars medians. Orange: a further run of LPJmL-FIT scored the same way, which must give about 1. (d) Signed area-total errors of stems and biomass. Dashed: limits of the pass standard.

run (limit 3, met in 7 of 15), biomass per tree 1.47 times, stem and biomass totals a median −5.2 % (−15 to +14 %) and −4.5 % (−23 to +9 %; limit 25 %). Of the 22 criteria of the pass standard, 18 are met: tree density fails in typical and worst-tenth dense cells and in typical sparse cells, biomass per tree in typical dense cells. With dense cells classed by the held-out run, as the standard was first written, 18 of 22 are also met: tree density then fails in typical dense cells (1.20, 7 of 15 cases) and typical sparse cells (3.15, 6 of 15), biomass per tree in typical dense cells (1.16, 9 of 15), and the sparse stem total is +20.0 % (11 of 15).

6.3 No drift in the forest state

Run year by year from 2020 to 2100 (Fig. 7), the emulator's error starts below that of a further run, because it knows the run's own 2000–2019 state, and rises over about three decades as that knowledge fades. It then levels off: biomass per tree at 1.02–1.06 times a further run's error in the 2050s and 1.09–1.14 in the 2090s, tree density at 1.12–1.15 in the 2050s and 1.20–1.24 in the 2090s. From 2051–2070 to 2081–2100 — the baseline chosen after the pre-registered one failed, see below — the typical error grows by a factor of 1.06 for tree density (all 15 cases below the limit of 1.2) and 1.05 for biomass per tree (13 of 15); the trait medians grow by 1.11–1.20 while remaining at 0.63–0.88 times a further run. The rise in the first decades does not depend on the scenario (tree

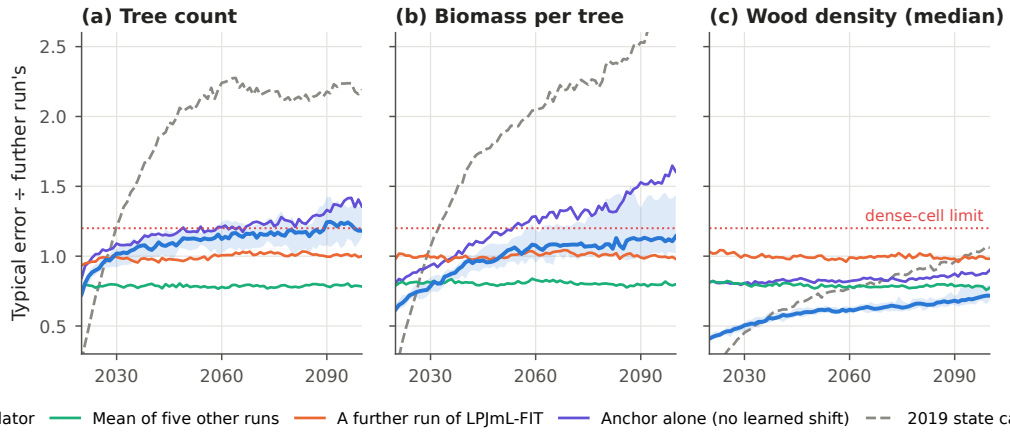


Figure 7. The forest maps run year by year: typical-cell error ratio of dense cells per year, median over the 15 held-out cases (shading: 10th–90th percentile of cases for the emulator), for (a) tree density, (b) biomass per tree and (c) median wood density. Reference arms: a further run (orange), the mean of five other runs (green), the anchor without the learned departure (violet) and the 2019 state carried forward (grey dashed).

density 1.22 under SSP1-2.6, 1.23 under SSP5-8.5), which identifies it as the fading start state rather than climate extrapolation. What remains is a slow creep of tree density in the second half of the century that increases with warming (+0.05, +0.10 and +0.12 from the 2050s to the 2090s for SSP1-2.6, SSP3-7.0 and SSP5-8.5). The reference arms behave as they should: a further run stays at 1.0, the mean of other runs at 0.8, the anchor alone rises steadily to 1.3–1.6, and the carried-forward 2019 state rises to more than twice a further run's error, so the measure does detect drift. The pre-registered drift rule, which compares 2081–2100 with 2021–2040, fails (1.22 for tree density, 1.41 for biomass per tree; 7 and 1 of 15 cases within the limit). We attribute this to the artificially low early error, which the scenario-independence of the early rise supports, but the rule as written is failed, and the late creep of tree density under strong warming is a real, if slow, growth of the gap.

6.4 Daily carbon and water exchange

At the three biome cells of Fig. 8 the full emulator — daily networks never trained on these cells, fed a stand from forest maps that never saw the climate model or the run — reproduces the seasonal course and the day-to-day variability of GPP and evapotranspiration: the spring onset in the temperate and boreal forests, the short wet season at the semi-arid cell, the response to individual dry spells. Year-to-year variations of annual GPP follow the original's at the temperate and semi-arid cells (correlation 0.93 and 0.94; a further run: 0.91 and 0.96), less so at the boreal cell (0.65; a further run: 0.99). The 2021–2100 means differ by +0.8 % at the temperate and −0.6 % at the boreal cell, but by +12 % at the semi-arid cell, where the original's own two runs differ by 3 %. The top-metre water store is the least faithful daily variable: at the temperate cell it dries 20 mm too deep in summer, and at the boreal cell it holds up to 50 mm more water in summer than the original, while the fluxes stay close. This is expected: the networks are trained on fluxes and store changes, and the store is a state integrated by the bucket, not a prediction target.

Over the 238 held-out cells and 13 legs (Fig. 9), the typical error of 30-year mean GPP is 2.4 times a further run's (median over legs; limit 3, met on 13 of 13 legs), evapotranspiration 3.1 times (4 of 13), NPP 3.9 times (0 of 13) and runoff 5.2 times. Since two runs of the original differ by only 1.0–1.6 % in a cell's 30-year GPP, evapotranspiration or NPP, this means typical per-cell errors of about 3–6 % for GPP and evapotranspiration and 5–6 % for NPP. Area totals are off by a median of 1 % for GPP, NPP and evapotranspiration and 2 % for runoff, and by at most 2.9 %

on any leg. The climate response (2071–2100 minus 2021–2050) is 1.8–2.4 times a further run's error, above the proposed 1.5. Runoff is hardest because it is a small residual of precipitation minus evapotranspiration, so the evapotranspiration error passes into it amplified; NPP is hard because it is a small difference of two large terms in the most productive cells.

The emulated stand costs accuracy. Fed the original's own stand instead, the same daily networks reach 2.2 (GPP), 2.6 (evapotranspiration), 3.4 (NPP) and 4.5 (runoff) times a further run's error. Compared leg by leg and period by period, the emulated stand adds a median 12 % to the GPP error and 37 % to the evapotranspiration error, most of it through the cover of the plant types. On these legs no flux drifts away from the original: the late-to-early ratio of single-year errors is 0.90–1.04 for the medians of all four fluxes, and only GPP on the two warmest legs (UKESM1-0-LL SSP3-7.0, IPSL-CM6A-LR SSP5-8.5) exceeds the limit (1.26 and 1.28). The deep water store stays within the range of the original's change, and over 81 years the across-cell mean of the top-metre store stays between 80 and 91 mm.

In this evaluation the climate models are seen by the daily networks; only the cells are new. That matters: in earlier tests with a climate model withheld from daily-network training entirely, the gap to the original in GPP and NPP grew by a factor of 1.3–1.4 over the century. The cause was the daily carbon response degrading as the climate moved away from the training climates, not the water stores, since a run with the stores reset daily from the original drifted identically. Adding further climate models to training did not reduce this growth beyond run-to-run noise for UKESM1-0-LL; it falls (to 1.17 for NPP) only when that model itself is in training, and even then the full emulator's GPP gap grows by 1.26 on that model's warmest leg. The daily networks therefore have not been shown to hold their accuracy under a climate model they have not seen.

6.5 Learning the warming response

The warming response is the property an ESM needs most and the one most emulators never test. We tested it on LPJmL-FIT with nitrogen limitation and rising CO_2, run on 674 cells that each represent one region of a 674-region climate classification of the land surface (25 patches each), under the ten climate models and three scenarios, with CO_2-fixed twins of SSP3-7.0 for the training climate models. Two climate models of low and high climate sensitivity, MPI-ESM1-2-HR and IPSL-CM6A-LR, were held out entirely, together with 20 % of the cells; both lie within the warming range of the eight training models, of which CanESM5 and UKESM1-0-LL warm more, so the test climate is covered by the training climates. An independent second run of the original exists for the held-out legs. Here the forest maps are anchored on each cell's own 1990–2019 state, so no future output of the original enters the prediction. The pass criteria were registered before the runs: on every held-out leg a global change of the same sign as and within ±25 % of the original's, a per-cell regression slope between 0.8 and 1.2, and a per-cell root-mean-square error at most 1.5 times the second run's.

With rising CO_2, the original's vegetation carbon increases by 32–45 % by 2071–2100 on the held-out legs (Fig. 10a), mostly through CO_2 fertilisation. The emulator reproduces 81–97 % of this change, while the original's independent second run reproduces 61–89 %: the original's own global response varies by up to 40 % between runs on this grid, because single cells stand for whole regions. Trained only on SSP1-2.6, the same maps reproduce 37–43 % — the failure seen when an emulator meets a climate warmer than its training climates.

With CO_2 fixed at its 2019 level, the original loses vegetation carbon under warming, by up to 20.7 % for IPSL-CM6A-LR under SSP5-8.5 (second run: 26.3 %). The maps as registered reproduce 67 % of the IPSL-CM6A-LR loss under SSP3-7.0 and 57 % under SSP5-8.5. After seeing this result we added CO_2-fixed twins of SSP1-2.6 and SSP5-8.5 to training, so that warming and CO_2 rise could be told apart for strong warming; the held-out models and cells were unchanged and nothing else was tuned. This raised both fractions to 72 %, short of the 75 % we had stated as the expectation for this step, and left the rising-CO_2 results unchanged within 0.01. Figure 10 shows this second version. The SSP1-2.6-only maps predict gains on all four warming-only legs,

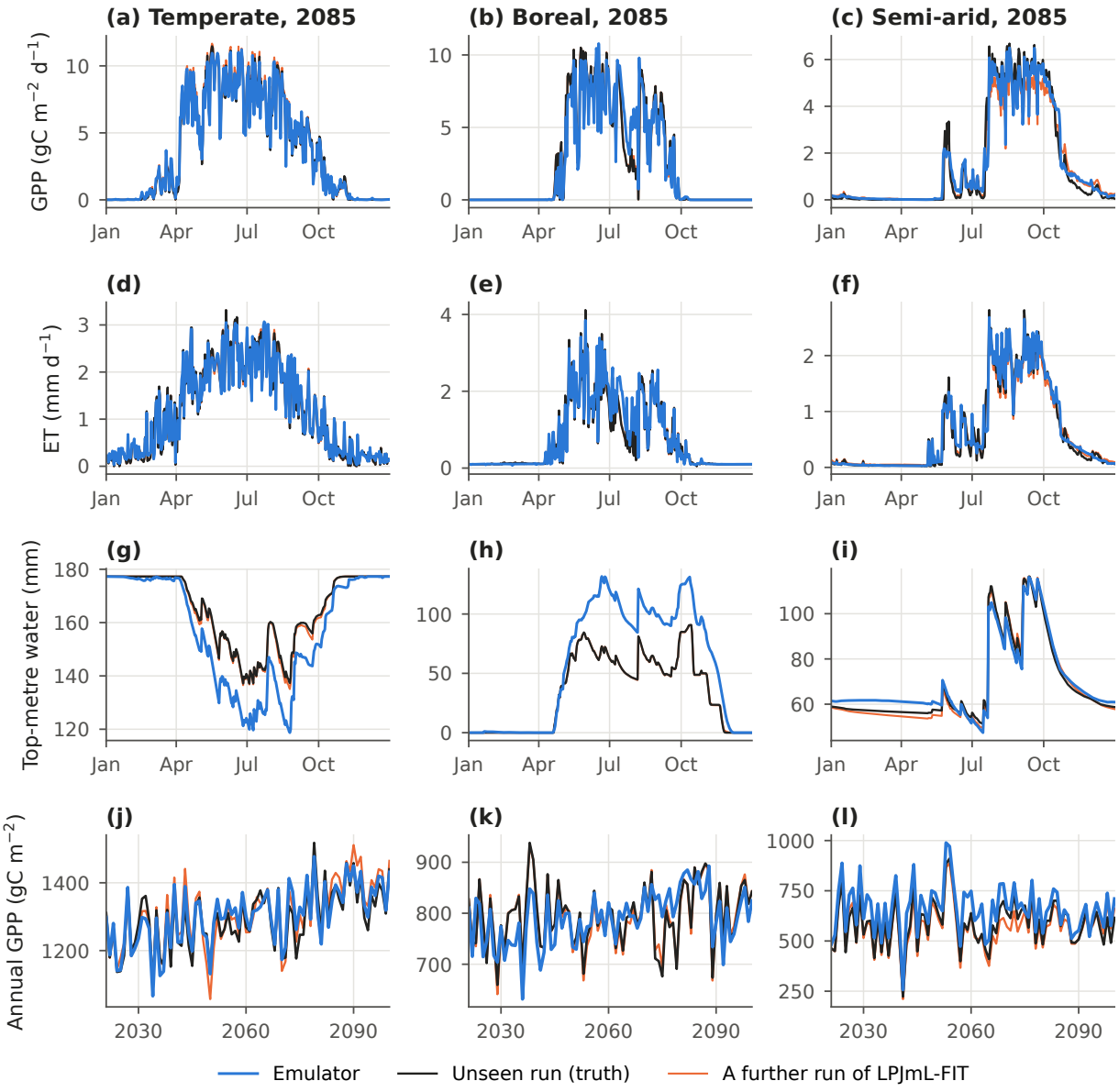


Figure 8. Daily output of the full emulator at three biome cells never used to train the daily networks (GFDL-ESM4, SSP3-7.0; forest maps with this climate model and run held out). (a–c) Daily GPP, (d–f) evapotranspiration and (g–i) top-metre soil water in 2085; (j–l) annual GPP 2021–2100. Black: the unseen run of LPJmL-FIT; orange: a further run; blue: the emulator.

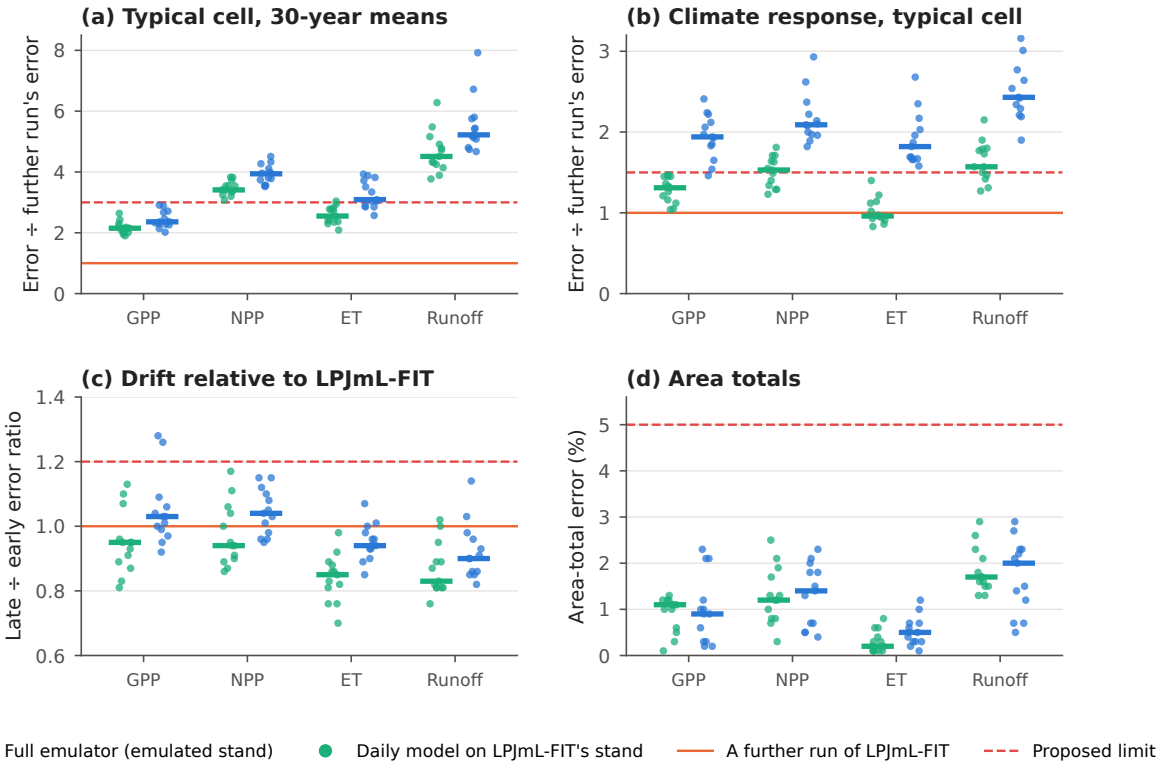


Figure 9. Daily fluxes over 238 held-out cells and 13 climate-model legs (dots: legs; bars: medians). (a) Typical-cell error ratio of 30-year means (larger of 2021–2050 and 2071–2100); (b) error ratio of the climate response; (c) drift, the late-to-early ratio of single-year errors (1 = no change in the gap to the original); (d) area-total errors. Blue: the full emulator; green: the same daily networks fed the original's stand. Orange line: a further run of the original; dashed: proposed limits.

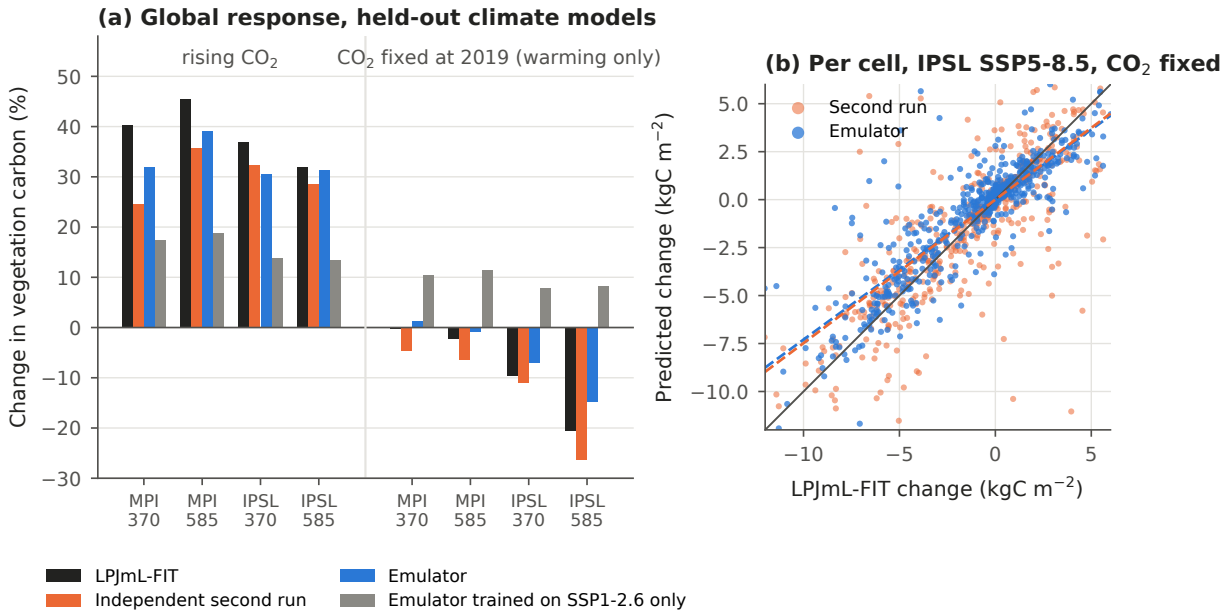


Figure 10. The warming response under two climate models never seen in training (LPJmL-FIT with nitrogen limitation, 674 representative cells). (a) Area-weighted change of vegetation carbon from 1990–2019 to 2071–2100 with rising CO_2 and with CO_2 fixed at 2019: the original, its independent second run, the emulator (the version with CO_2-fixed twins of every training scenario, added after the first held-out results; see text), and the same emulator trained on SSP1-2.6 only. (b) Per-cell change for IPSL-CM6A-LR, SSP5-8.5, fixed CO_2: emulator (blue) and second run (orange) against the original; dashed lines are least-squares slopes, the solid line is the 1:1 line.

because all their training legs carry rising CO_2. Per cell (Fig. 10b) the emulator's change is as close to the original's as the second run's is (root-mean-square error 0.65–0.80 times the second run's), and the slopes of both against the original are below one (0.73 and 0.75): the original's per-cell change contains run-to-run noise, which flattens the regression slope of any predictor, including a perfect estimate of the expected response.

For GPP the picture splits. The daily networks fed the original's stand reproduce the global GPP change within 4 % and the per-cell change with slopes of 0.92–0.99, with rising and with fixed CO_2. Fed the emulated stand, the global change with rising CO_2 is still within 6 %, but the per-cell slope falls to 0.66–0.79, because the forest maps' response of leaf area and tree cover is too uniform across cells (slopes 0.73–0.83, against 0.89–0.96 for the second run). The global totals are right; the cell-to-cell spread of the response is too small. On the held-out cells, a learning curve over 135, 269 and 539 training cells reduced this error by 6–12 % per doubling, which points to the number of training cells rather than the model form as the limit; extrapolated, reaching 1.2 times a second run for leaf area would take about 12 000 training cells, the scale of a full-grid training set.

Judged by the registered criteria, the test was *not passed*. The global changes and the root-mean-square errors pass on every leg. The slope criterion fails for vegetation carbon on three of four rising-CO_2 legs — a criterion that the original's own second run also fails (slopes 0.72–0.84), for the reason given above — and for GPP fed the emulated stand on all four, which is a real defect. With CO_2 fixed, the largest warming-only loss stays damped by about 30 %.

6.6 Full-grid application

To show that the emulator runs globally, we retrained the forest maps on four runs of LPJmL-FIT (nitrogen off) under GFDL-ESM4 — historical, SSP1-2.6, SSP2-4.5, SSP3-7.0 — made with a different build of the model, anchored each cell on its own simulated 1985–2014 state, and emulated each future scenario with maps that never saw it. The daily networks were not retrained

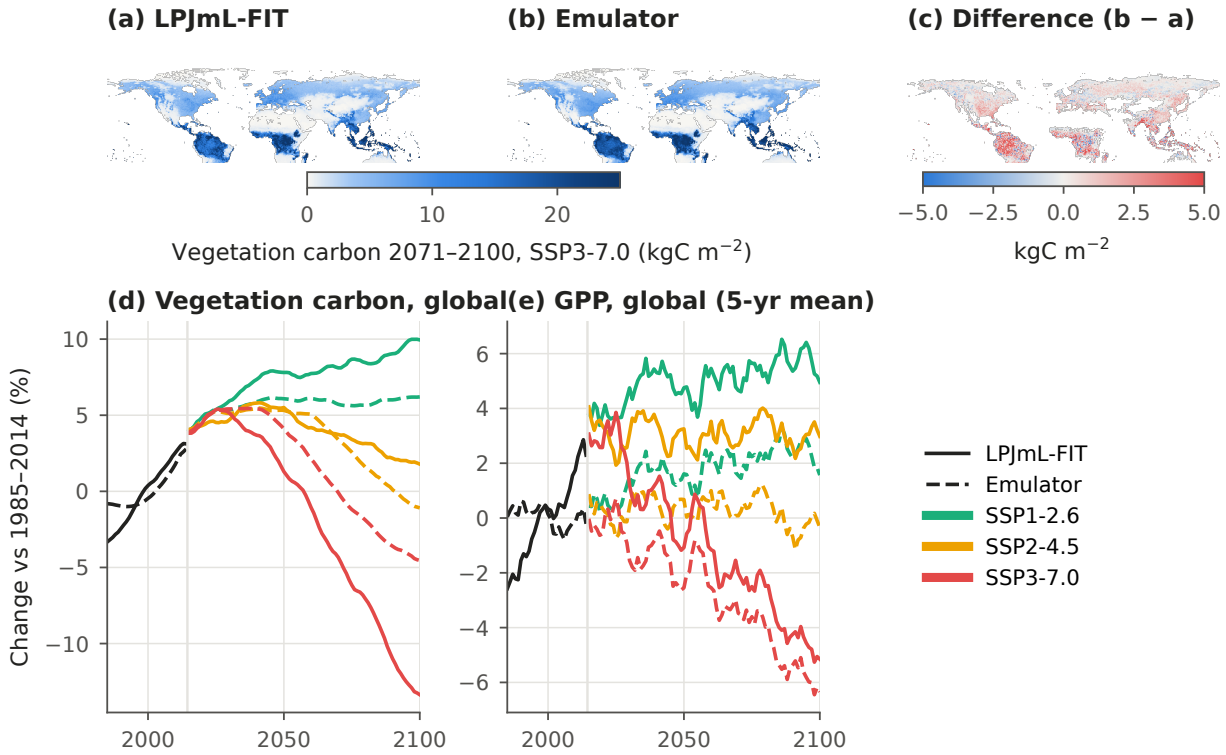


Figure 11. LandEmu on the full 0.5° grid. Vegetation carbon in 2071–2100 under SSP3-7.0 in (a) LPJmL-FIT and (b) the emulator, whose forest maps never saw SSP3-7.0, and (c) their difference (cells with at least 0.5 kgC m^{-2}). (d) Global vegetation carbon and (e) global GPP as change relative to the original's 1985–2014 mean (5-year running means): LPJmL-FIT solid, emulator dashed. The historical forest state is anchored and therefore not a test.

(these runs store no daily output). LandEmu then ran on all 64 240 land cells from 1975 to 2100 (Fig. 11). Over 1985–2014 the emulated global GPP equals the original's (+0.04 %), NPP is 1.1 % high, transpiration 8.5 % low and runoff 5.8 % high. Benchmarked against observations with the ILAMB system, the emulator's overall scores are within 0.03 of the original's on all 27 data-set comparisons; for biomass, leaf area and cover this is not a test of the emulator, because its historical forest state is anchored on the original's. Over the same period the original's GPP rises from 151 to 162 PgC yr^{-1} because CO_2 rises in its historical forcing, while the emulator's stays flat: the nitrogen-off emulator has no CO_2 input by design (Sect. 2), so it reproduces the level but not this trend.

Under the unseen scenarios, the typical per-cell error in 2071–2100 is 6.0–6.4 % for GPP, 4.4–4.8 % for evapotranspiration, 11–15 % for vegetation carbon and 7–8 % for dense-cell tree density, always smaller than when the 1985–2014 state is carried forward (11–15 %, 7–12 %, 18–26 % and 10–11 %). The global responses agree in sign in 13 of 15 cases (five quantities, three scenarios), but the carbon responses are 3–5 percentage points too low, and under SSP3-7.0, the warmest scenario and the only one outside the training range, the emulator captures only a third of the original's vegetation-carbon loss (−2.8 against −9.2 %). This is the extrapolation failure of Sect. 6.5 reproduced on the full grid: with only milder scenarios in training, the strongest response is damped. On an earlier set of eight independent runs of this model under GFDL-ESM4, scored over all 57 726 tree-bearing cells against a further run, an earlier form of the forest maps that predicts 30-year windows rather than single years passes the dense-cell standard when only the run is held out (tree density and biomass per tree 0.92 times a further run) but not when the scenario is held out as well (1.24 and 1.47 times).

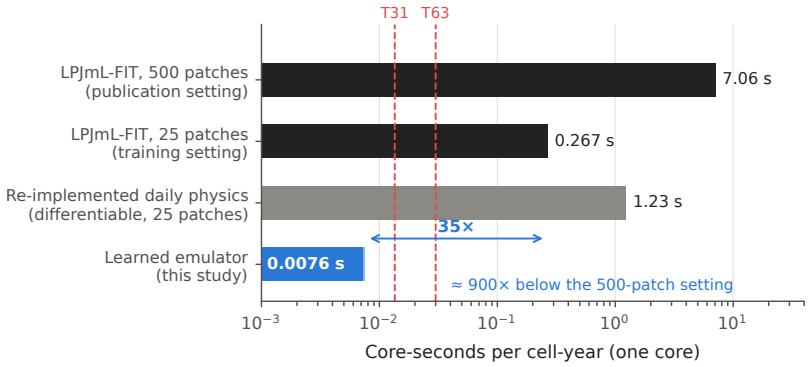


Figure 12. Cost per cell-year on one core (marginal rate). LPJmL-FIT at its training and publication settings, the differentiable re-implementation of its daily physics, and LandEmu (dark blue: daily networks, 97 %; light blue: yearly statistics and forest maps). Dashed: land allowances for T31 and T63 atmospheres (10 % of a coupled atmosphere's cost).

6.7 Cost

Measured end to end on one core over 1 050 cells and 81 years, the whole emulator costs 7.6×10^{-3} core-seconds per cell-year (Fig. 12), 97 % of it in the daily networks. LPJmL-FIT, timed on one core of the same computing cluster at the temperate cell (marginal rate), costs 0.267 core-seconds at 25 patches and 7.06 at 500 patches, so the emulator needs 35 times fewer core-seconds than the original at its training setting and about 900 times fewer than at the publication setting. The original's cost depends on how many trees a cell carries: across the five biome cells it is 0.20–0.33 core-seconds at 25 patches, which gives factors of 26–44. The re-implemented differentiable physics of the first design cost 1.23 core-seconds. The emulator lies inside the allowance for an intermediate-resolution atmosphere by a factor of 1.8 (T31) to 4 (T63). The nitrogen-on variant costs 7.3×10^{-3} core-seconds, 115–130 times less than that model on the same cells. Training is cheap as well: about 15 GPU-minutes per network and seconds to minutes per forest map. The cost is dominated by one matrix product per day and network, so batching over years or a GPU would reduce it further; neither is used here.

7 Discussion

7.1 LandEmu against the requirements for ESM-ready land emulators

Table 2 compares LandEmu with the six requirements derived in the review (Priesner, 2026). Three design choices carry most of the result. *Learning the whole daily step* is what makes the emulator fast: the literature's large speed-ups come from replacing an expensive iterative solver or a whole model step (Wesselkamp et al., 2025; Yang et al., 2026; Hemmen et al., 2026), and our profile of LPJmL-FIT shows that nothing short of this can give an order of magnitude. Re-implementing physics can be faster than legacy code when the code is a column model without a per-individual loop (Lahlou et al., 2026); for an individual-tree model it was not. *Predicting the expectation* removes the patch ensemble, which is a variance-reduction device of the original, not a process; at the publication setting it is worth more than every process combined. *Keeping the forest state non-recursive* sidesteps the error accumulation that made our one-step demography model unusable and that rollout losses (Raoult et al., 2026; Wang et al., 2023) or trajectory fitting (Ekholm et al., 2024) address in other emulators. The daily stores are recursive, but they are closed physical stores rather than a learned latent state, and they did not walk away from the original in 81-year free runs.

Table 2. LandEmu against the six requirements for ESM-ready land emulators of Priesner (2026).

Requirement	LandEmu v1
Prognostic state and memory	explicit snow, top-metre and deep water stores; trailing weather windows; 30-year climate window; stand carried yearly. No learned latent state. Forest state not recursive by design.
Coupling interface	daily latent-heat-relevant fluxes (interception, transpiration, soil evaporation), runoff, snow; GPP and NPP; leaf area and plant-type cover from which albedo and roughness can be derived. Missing: energy fluxes and surface temperature (LPJmL-FIT computes none), heterotrophic respiration and fire, hence the net CO_2 flux.
Conservation	water closed exactly every day by construction. Carbon is not tracked between NPP and vegetation carbon, which are predicted separately.
Stability under feedback	structure error grows by a factor of 1.05–1.06 after the start state fades, with a slow creep of tree density under strong warming (the pre-registered drift rule fails); no flux drift on climate models seen in training except GPP on the two warmest legs; day-by-day interface identical to the yearly call. No feedback with a host model tested here.
Out-of-distribution climate	yearly maps: climate model and independent run held out; daily networks: places held out, climate models seen (with one withheld, the GPP and NPP gap grew 1.3–1.4-fold); warming response tested on unseen climate models with and without CO_2 rise, not passed as pre-registered; extrapolation beyond the training scenarios fails (one third of the loss).
Uncertainty, interfaces, openness	errors expressed relative to the original's run-to-run variability; deterministic prediction of the expectation, no predictive distribution; numpy + LightGBM package with daily and yearly calls.

7.2 What limits the fidelity

Three findings locate the remaining error. First, the emulator is *data-limited in the space of climates, not in the number of runs.* On the panel, adding five climate models to training raised the share of cells close to the unseen run by 0.023 and lowered the biomass-per-tree error ratio from 1.64 to 1.42, while adding two runs of the original raised it by only 0.005. The warming-response test makes the same point from the other side: with training climates that cover the test climate the response is reproduced, with only milder climates it is damped by more than half. Second, *sparse stands* carry most of the structure error. Reweighting the loss towards them (logarithmic targets, Poisson and Tweedie objectives) only moved error between sparse and dense cells, while the mean of three runs is 0.82 times a further run in every density class — so their response is predictable in principle, and what is missing is information about how sparse cells respond. Third, the daily networks' own error, two to three times a further run, still exceeds the 10–40 % that the emulated stand adds, but the stand's share is concentrated in evapotranspiration and in the cell-to-cell spread of the climate response.

7.3 Limitations

- **The original must have been run on every emulated cell.** The forest maps depart from a per-cell anchor. In the panel version the anchor is the original's mean over its training runs for each year to 2100; in the global version it is the cell's simulated 1985–2014 state. Either way, a cell never simulated by LPJmL-FIT cannot be emulated, and on the response test the error on the 20 % of held-out cells is 1.8–2.2 times a second run's for leaf area and cover, compared with near-parity on training cells.
- **The cells are seen in training.** The structure tests hold out the climate model and the run, not the place. This mirrors deployment on the cells the original was run on; with cells held out as well, the panel scores were clearly worse.

- **Daily NPP and runoff** miss the proposed limit (3.9 and 5.2 times a further run), and the climate response of every flux is 1.8–2.4 times a further run's error.
- **The daily networks have seen every climate model they were scored on.** With a climate model withheld, their GPP and NPP gap to the original grew 1.3–1.4-fold over the century.
- **Some evaluation choices were made with results known**: classing dense cells by the anchor, the 2051–2070 drift baseline for the forest structure, the limits for the daily fluxes, and the CO_2-fixed training runs of the response test. We report the outcome under the pre-registered choices beside each.
- **Extrapolation.** Under a scenario warmer than all training scenarios the response is damped (Sect. 6.6); deployment should keep the training climates broader than the target climate.
- **Scope of the outputs.** The structure targets cover trees taller than 5 m, as the original's standard output does, and the traits are represented by their medians, not their full distributions. There are no energy fluxes and no net CO_2 flux.
- **Scale of the evaluation.** The detailed evaluation uses 1 050 of 54 020 tree-bearing cells; the full-grid application has one run per scenario, so its errors cannot be set against a further run.

7.4 Outlook

The day-by-day interface makes LandEmu callable from an atmosphere model; experiments with Earth system models will be reported separately. The most direct improvements are training on a full-grid set of runs that covers warmer climates (which removes the unseen-cell limit and the extrapolation problem together), a stand map aimed directly at the daily networks' inputs, explicit litter and soil carbon pools so that the net CO_2 flux is complete, and a calibrated predictive spread scored against several independent runs of the original.

8 Conclusions

LandEmu v1 emulates an individual-tree, trait-based DGVM as a whole: its daily carbon and water exchange and its yearly forest structure, trait composition and stand, from daily weather alone, with water conserved exactly. Its design — learn the whole daily step, predict the expectation over the patch ensemble, keep the forest state non-recursive, train on climate contrast and judge against the original's own variability — followed from measurements that ruled out a hybrid of learned demography and re-implemented physics. Under a held-out climate model and run, dense forests are reproduced near the original's run-to-run variability but just outside the limit set for them (tree density 1.26, biomass per tree 1.18 times a further run, limit 1.2; trait medians better than a further run), and the structure error grows only slowly once the start state has faded. On places held out from their training, the daily networks reproduce GPP and evapotranspiration to a few per cent per cell, without drift on the climate models they were trained on; under an unseen climate model their carbon fluxes drift. Trained on climates that cover the test climate, the emulator reproduces the change in vegetation carbon under rising CO_2 as well as a second run of the original does on the global total, but it damps the largest warming-only losses, the cell-to-cell spread of the response and the response of sparse stands, and it did not pass the test as pre-registered. At 7.6×10^{-3} core-seconds per cell-year it costs 35–900 times less than the original and fits the land budget of an intermediate-resolution Earth system model.

Code and data availability

The emulator package (`python/src/lpjmlfit_emulator/landemu`), the training, scoring and release scripts, the scripts that produce every figure of this paper (`scripts/paper`) and the decision records behind every number are in the project repository `https://github.com/rimajj/LPJmLFIT_Emulator` (to be archived on Zenodo on acceptance). The LPJmL-FIT simulations

and the released emulator (models and per-cell set-ups) are stored at the Potsdam Institute for Climate Impact Research and will be archived with the code.

Author contributions

JP conceived the study, defined its goals and acceptance standards, and made the design decisions. The implementation, simulations, analysis, figures and drafting were carried out by an AI research assistant (Claude, Anthropic) under JP's direction, with every decision recorded in the repository; JP is responsible for the final content.

Competing interests

The author declares no competing interests.

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